

Week 5 supervised learning

1. what is a classifier and labels?

A classifier predicts/maps each object or datapoint to a unique label. Label is a category or group.

2. What is binary and multiclass classification?

Binary only has two classes/labels. Multiclass has more than two

3. What are features?

Features are representations of an object. We have to decide what features we want to observe and the possible values. Then we have to extract the values for each object. For example a flower can have height and width of the petal.

4. What algorithms can only take numerical features?

Perceptron, KNN, linear regression, logistic regression.

5. What algorithms can only take categorical features?

Decision trees

6. What is multi-label classification?

Multi-label classification is when an object has more than one label.

7. What are the stages in supervised learning?

Extract features from input. Train the data on training set, predict the test set.

8. What is confusion matrix? What is the goal?

Confusion matrix evaluates the classification algorithm. The goal is to evaluate the classifier. The matrix contains true positive, true negative, false positive and false negative.

9. How do we calculate accuracy?

True positive+ true negative / size of dataset

10. How do the k nearest neighbor algorithm work? What are the stages? How do we choose k and what factor decides the value of k? what are the downsides of KNN

kNN works by calculating the distance from the point we want to find to all the other points. The most common way to calculate distance is the Euclidian. Then we choose the k nearest. Choose the majority class for this set.

Choose the K that gives the best accuracy. We can find the accuracy by train on different values of k. too small k can risk too good of a fit to the training data, which can lead to overfitting. Too large k can make the algorithm too general.

The downside of kNN is that we have to calculate the distance for all the points, which is labor-consuming and inefficient.

11. Why do we scale the data? Should we apply scaling/normalizing before or after splitting it into training and testing?

Feature scaling is a preprocessing technique used in supervised learning to standardize or normalize the input features of a dataset. The goal is to bring all features to a similar scale, preventing certain features from dominating the learning process due to differences in their magnitudes.

We scale the data because it improves model performance, faster convergence and easier interpretation. It can lead to better efficiency, but it is not guaranteed that the results are better.

Split the data first into training and testing, then apply scaler to training. Then scale the data that you want to predict.

12. What is max-min-scaling?

Finds the minimum and maximum value for each feature . the scale_i = $(x_{j,i} - \min_i) / (\max_i - \min_i)$

13. What is normalizing?

Finds the mean and standard deviation for each feature. The function is scale_i = $(x_{j,i} - \text{mean}_i) / \text{std}_i$